

Applications of the definite integral

Areas between curves, volumes of revolution, distance from velocity — one integral, four questions.

LESSON 52–54

3 × 40 MIN

ALGEBRA 11, §3.2

P1 · 10.5–10.7

LESSON CLOCK

20'	30'	30'	26'	30'	9'
-----	-----	-----	-----	-----	----

Area with the axis 20 min

Area between two curves 30 min

Volumes of revolution 30 min

Motion 26 min

Practice 30 min

Homework 9 min

LEARNING OBJECTIVES

- Find the area between a curve and the x-axis, taking sign into account.
- Find the area between two curves.
- Find a volume of revolution about the x-axis.
- Recover distance from velocity and velocity from acceleration.

TEXTBOOK

National pp. 243–260

Cambridge Pure Mathematics 1, pp. 227–240

Workbook P1 Exercise 10E, 10F

01 Explanation

Area with the axis

THE ROUTINE

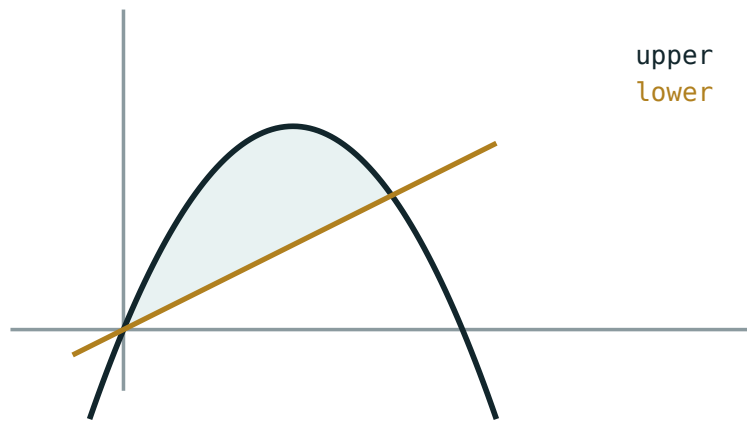
1 Find where the curve meets the axis inside the interval. **2** Integrate over each piece. **3** Add the absolute values.

Skipping step 1 is the standard error: a curve that dips below the axis returns a partial cancellation instead of an area.

$$\text{area} = \left| \int_a^c f \right| + \left| \int_c^b f \right| \text{ where } f(c) = 0$$

Area between two curves

$$\text{area} = \int_a^b (\text{upper} - \text{lower}) dx$$



$$\text{area} = \int (\text{upper} - \text{lower}) \, dx$$

Upper minus lower, integrated between the two crossing points.

The limits a and b are the x -coordinates where the curves cross, found by solving $f(x) = g(x)$ first.

Example. $y = 2x$ and $y = x^2$ meet at $x = 0$ and $x = 2$. Between them the line is above:

$$\int_0^2 (2x - x^2) \, dx = \left[x^2 - \frac{x^3}{3} \right]_0^2 = 4 - \frac{8}{3} = \frac{4}{3}$$

WHY THE SIGN TAKES CARE OF ITSELF

Because $\text{upper} - \text{lower}$ is positive throughout, the integral is the area directly — even if both curves are below the axis. That is why this method is safer than integrating each separately.

Volumes of revolution

Rotate the region under $y = f(x)$ between a and b about the x -axis. Each thin strip becomes a disc of radius y and thickness dx :

$$V = \pi \int_a^b y^2 \, dx$$

CURVE	INTERVAL	SOLID	VOLUME
$y = r$	$[0, h]$	cylinder	$\pi r^2 h$
$y = \frac{r}{h} x$	$[0, h]$	cone	$\frac{1}{3} \pi r^2 h$
$y = \sqrt{r^2 - x^2}$	$[-r, r]$	sphere	$\frac{4}{3} \pi r^3$

Every volume formula of Grade 11 geometry is one of these integrals. The $\frac{1}{3}$ in the cone is $\int x^2 dx$; the $\frac{4}{3}$ in the sphere is $\int (r^2 - x^2) dx$ over $[-r, r]$.

SQUARE THE y , DO NOT SQUARE THE INTEGRAL

$\pi \int y^2 dx$, not $\pi (\int y dx)^2$. The two differ completely.

Motion

$$s = \int v dt \quad v = \int a dt$$

Integration undoes the differentiation of Quarter I. With limits it gives a definite answer:

$$\text{displacement} = \int_{t_1}^{t_2} v dt \quad \text{distance} = \int_{t_1}^{t_2} |v| dt$$

DISPLACEMENT IS NOT DISTANCE, AGAIN

If v changes sign in the interval, split at that instant and add the absolute values — exactly as for area. It is the same theorem wearing different words.

02 Worked examples

EXAMPLE 1 Find the area between $y = 2x$ and $y = x^2$.

① $2x = x^2 \Rightarrow x = 0, 2$

The limits.

② The line is above between them.

③ $\int_0^2 (2x - x^2) dx$

④ $= 4 - \frac{8}{3} = \frac{4}{3}$

ANSWER

$$\frac{4}{3}$$

EXAMPLE 2 The region under $y = \sqrt{x}$ from 0 to 4 is rotated about the x-axis. Find the volume.

① $y^2 = x$

② $V = \pi \int_0^4 x dx$

③ $= \pi \left[\frac{x^2}{2} \right]_0^4$

④ $= 8\pi$

ANSWER

$$8\pi \approx 25.1$$

EXAMPLE 3 A body has $v = 3t^2 - 12t$ m/s. Find the distance travelled in the first 5 s.

① $v = 3t(t - 4)$, zero at $t = 0, 4$.
Split at 4.

② $\int_0^4 v \, dt = [t^3 - 6t^2] = -32$

③ $\int_4^5 v \, dt = (125 - 150) - (-32) = 7$

④ $32 + 7$

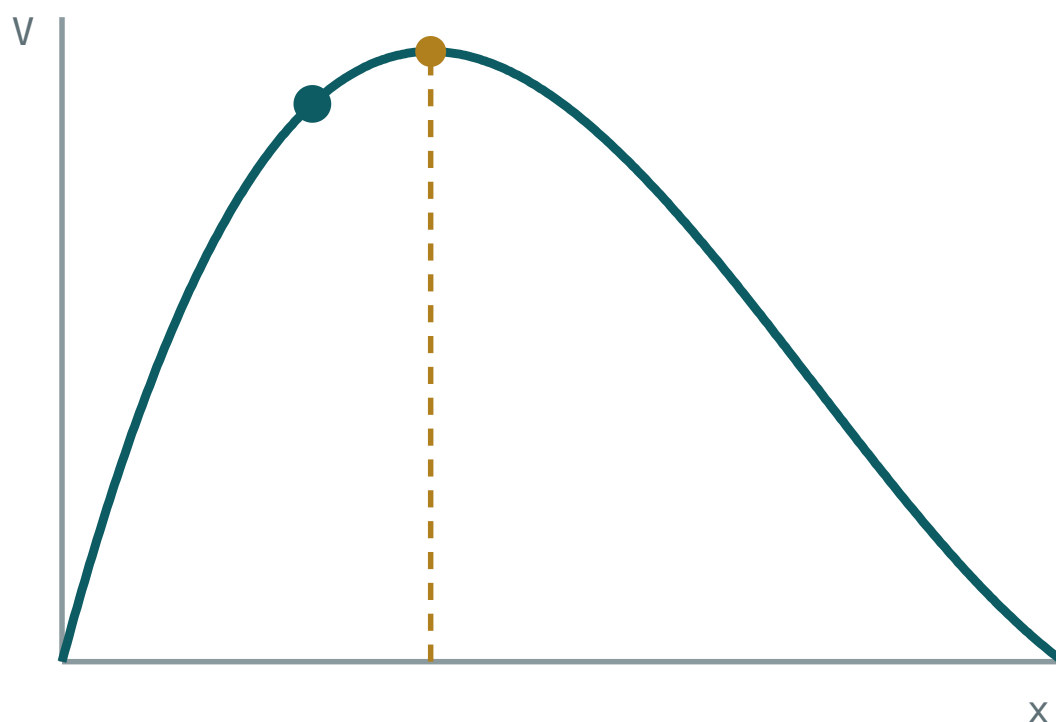
ANSWER

39 m

03 Interactive model

Sketch the two curves and shade the region before writing any integral.

Area under a curve


 $x = 2$
 $V(x) = 384$

greatest $V = 420.1$

at $x = 2.94$

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Area between curves	Egri chiziqlar orasidagi yuza	Площадь между кривыми
Upper and lower curve	Yuqori va quyi egri chiziq	Верхняя и нижняя кривая
Point of intersection	Kesishish nuqtasi	Точка пересечения
Volume of revolution	Aylanish jismi hajmi	Объём тела вращения
Axis of revolution	Aylanish o'qi	Ось вращения
Distance travelled	Bosib o'tilgan yo'l	Пройденный путь
Displacement	Ko'chish	Перемещение
Signed area	Ishorali yuza	Площадь со знаком
Integrand	Integral ostidagi funksiya	Подынтегральная функция

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The area between two curves is:

$$\int f + \int g$$

$$\int(\text{upper} - \text{lower})$$

$$\int |f|$$

$$\int f \cdot g$$

The limits come from:

the question always

solving $f = g$

the axis

guessing

A volume of revolution about Ox is:

$$\pi \int y \, dx$$

$$\pi \int y^2 \, dx$$

$$(\pi \int y \, dx)^2$$

$$\int y^2 \, dx$$

Distance from velocity is:

$$\int v \, dt$$

$$\int |v| \, dt$$

$$v'$$

$$\int v^2 \, dt$$

A curve dipping below the axis needs:

nothing special

splitting at the zeros

squaring

doubling

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Area under $y = x$ from 0 to 4

ANSWER 8

2. Area under $y = x^2$ from 0 to 3

ANSWER 9

3. Area under $y = 2$ from 1 to 5

ANSWER 8

4. Volume from $y = 2$ on $[0, 3]$ about Ox

ANSWER 12π

5. Volume from $y = x$ on $[0, 3]$

ANSWER 9π

6. $v = 4$; distance in 6 s

ANSWER 24 m

7. $v = 2t$; distance in 3 s

ANSWER 9 m

Medium · the standard question

7 PROBLEMS

1. Area between $y = 2x$ and $y = x^2$

ANSWER $\frac{4}{3}$

2. Area between $y = x$ and $y = x^2$

ANSWER $\frac{1}{6}$

3. Area between $y = 4 - x^2$ and the axis

ANSWER $\frac{32}{3}$

4. Volume from $y = \sqrt{x}$ on $[0, 4]$

ANSWER 8π

5. Volume from $y = x^2$ on $[0, 2]$

ANSWER $\frac{32\pi}{5}$

6. $v = 3t^2 - 12t$; distance in the first 5 s

ANSWER 39 m

7. Same; the displacement

ANSWER -25 m

Hard · stretch and reason

7 PROBLEMS

1. Area between $y = x^2$ and $y = x^3$

ANSWER $\frac{1}{12}$

2. Area between $y = x^2 - 4$ and $y = 5$

ANSWER 36

3. Volume from $y = \frac{r}{h}x$ on $[0, h]$

ANSWER $\frac{1}{3}\pi r^2 h$

4. Volume from $y = \sqrt{r^2 - x^2}$ on $[-r, r]$

ANSWER $\frac{4}{3}\pi r^3$

5. Area enclosed by $y = x^3 - x$

ANSWER $\frac{1}{2}$

6. $a = 6t$, $v(0) = 0$, $s(0) = 2$. Find $s(3)$

ANSWER 29

7. Volume from $y = \frac{1}{x}$ on $[1, 3]$ about Ox

ANSWER $\frac{2\pi}{3}$

07 Homework

Homework — 6 tasks

Sketch and shade before every integral; find the limits before writing them.

- Find the area between $y = 3x$ and $y = x^2$.
- Find the area between $y = 9 - x^2$ and the x -axis.

3. Find the volume when the region under $y = x$ on $[0, 4]$ is rotated about Ox .
4. Find the volume when the region under $y = \sqrt{4 - x^2}$ on $[-2, 2]$ is rotated about Ox , and name the solid.
5. A body has $v = t^2 - 4t$ m/s. Find the displacement and the distance in the first 6 s.
6. Explain in three sentences why “upper minus lower” needs no absolute value.